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Department of Computer Science and Engineering

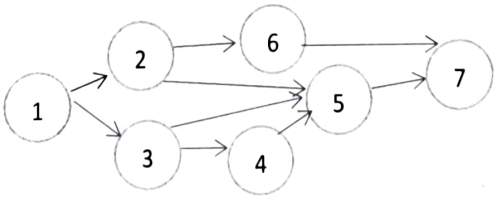
Academic year 2024-2025 (Odd Semester)

CIE II: QUESTION PAPER (Retest)

Course	ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING		
Date	February 2025	Maximum Marks	60
Course Code	IS353IA	Duration	110 Min
Sem	V	CIE – II	

Retest

SL No.	Part A	M	BT	CO
1.1	In BFS, nodes are expanded level by level, and it is guaranteed to find the shallowest solution if the branching factor is finite.	1	2	2
1.2	Uninformed search strategies do not use problem-specific information, whereas informed search strategies rely on heuristic to guide the search.	1	1	2
1.3	What is the role of the parameter k in the K-Nearest Neighbour (KNN) algorithm, and how does it affect the model's performance? balance b/w noise & generalisation	2	2	1
1.4	What assumption does the Naive Bayes classifier make about the features?	2	2	3
1.5	Given the logistic regression model logit(P)= $-2 + 0.5X_1 + 1.5X_2$, where X_1 and X_2 are predictor variables, calculate the probability P when $X_1=2$ and $X_2 = 1$.	2	3	1
1.6	Logistic regression uses the sigmoid function to map predicted values to probabilities, while Naive Bayes calculates probabilities using bayes theorem.	2	1	2

	Part B			
1.a	Discuss various criteria to measure the performance of search algorithms. Provide the comparative study of the uninformed search algorithms based on the performance criteria.	10	3	4
2.a	Given the weighted graph below and heuristics mentioned, determine the optimal path and cost to reach the goal node(7) from the initial node (1) using A* search algorithm.  cost(1,2)=4, cost(1,3)=3, cost(2,6)= 5, cost(2,5)=12, cost(3,4)=7, cost(3,5)=10, cost(4,5)= 2, cost(5, 7)= 5, cost(6,7)=16. h(1)=14, h(2)=12, h(3)=11, h(4)=6, h(5)=4, h(6)=11.	6	4	3

2.b	Write the algorithm for the k-nearest neighbor classifier	4	1	3
3.a	Explain the working of the Naive Bayes classifier with an example. Highlight its key assumptions, advantages, and limitations.	10	2	2
4.a	Differentiate between probabilistic and discriminative models	4	2	1
4.b	How do logistic regression and Naive Bayes classifier differ in their approach to classification?	6	3	2
5.a	Explain the concept of logistic regression. Discuss its mathematical formulation, working, advantages, limitations, and use cases with an example.	10	2	3

Course Outcomes: After completing the course, the students will be able to:-	
CO 1	Explain and apply AI and ML algorithms to address various requirements of real-world problems.
CO 2	Design and develop AI and ML solutions to benefit society, science, and industry.
CO 3	Use modern tools to create AI and ML solutions.
CO 4	Demonstrate effective communication through team presentations and reports to analyze the impact of AI and ML solutions on society and nature.
CO5	Conduct performance evaluation, modeling, and validation of AI and ML solutions benefiting lifelong learning.

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Particulars	CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
Marks Distribution	8	20	22	10	-	7	29	18	6	-	-